

BIO-DATA

1. Name and full correspondence address:
Dr. Shelaka Gupta
Assistant Professor
Department of Chemical Engineering
Indian Institute of Technology Hyderabad
Kandi Sangareddy Telangana 502285

2. Academic Qualification (Undergraduate Onwards)

	Degree	Year	Subject	University/Institution	% or GPA
1.	Ph.D	2014-2019	Chemical Engineering	IIT Delhi	NA
2.	M.Tech	2011-2013	Chemical Engineering	IIT Roorkee	9.06/10.00
3.	B.Tech.	2007-2011	Chemical Engineering	DCRUST Murthal, Sonapat, Haryana	84.6%

3. Ph.D. thesis title, Guide's Name, Institute/Organization/University, Year of Award
Catalytic Conversion of Biomass-Derived Platform Molecules: Mechanistic Insights, Fundamental Challenges and Opportunities for Rational Catalyst Design
Prof. M. Ali Haider
Indian Institute of Technology Delhi, 2019

4. Work experience (in chronological order)

S.No.	Positions held	Name of the Institute	From	To
1	Postdoctoral Research Fellow	IIT Delhi	01/04/2019	16/08/2019
2	Assistant Professor	BITS Pilani	19/08/2019	02/2020
3	Assistant Professor	IIT Hyderabad	2/03/2020	Present

5. Professional Recognition/ Award/ Prize/ Certificate, Fellowship received by the applicant.

S.No.	Name of Award/Recognition	Awarding Agency	Year
1.	Teaching Excellence Award	IIT Hyderabad	2024
2.	Selected as one of the 75 Women in STEAM and featured in the second edition of She Is	Govt. of India	2022
3.	Sumant Sinha Sustainability Leadership Award	IIT Delhi	2020
4.	FITT Best Industry Relevant Thesis Award	IIT Delhi	2019
5.	Gandhiyan Young Technology Innovation (GYTI) Award	SRISTI-BIRAC	2018
6.	Bioenergy Awards for Cutting Edge Research	DBT and IUSSTF	2017

6. Publications (List of papers published in SCI Journals, in year wise descending order).

Section A: List of publications before joining IITH.

	Publication Title	Authors	Journal Name	Page No.	Volume	Year	I.F.	Quartile
1.	Mechanistic Insights into the Pathways of Phenol Hydrogenation on Pd Nanostructures	Govind Porwal, Shelaka Gupta, S. Sreedhala, Joes Elizabeth, Tuhin Suvra Khan, M. Ali Haider and C. P. Vinod	ACS Sustainable Chemistry and Engineering	17126-17136	7	2019	7.3	Q1
2.	Mechanistic Approaches Towards Rational Design of a Heterogeneous Catalyst for Ring-Opening and Deoxygenation of Biomass-Derived Cyclic Compounds	Shelaka Gupta, Md. Imteyaz Alam, Tuhin S. Khan and M. Ali Haider	ACS Sustainable Chemistry and Engineering	10165-10181	7	2019	7.3	Q1
3.	Synergistic Effect of Zn in a Bimetallic PdZn Catalyst: Elucidating the Role of Under-coordinated Sites in the Hydrodeoxygenation Reactions of Bio-renewable Platforms	Shelaka Gupta, Tuhin S Khan, Basudeb Saha and M. Ali Haider	Industrial & Engineering Chemistry Research	16153 – 16163	58	2019	3.9	Q1
4.	On the Role of Oxocarbenium Ions Formed in Brønsted Acidic Condition on γ -Al ₂ O ₃ Surface in the Ring-Opening of γ -valerolactone	Tuhin S. Khan, Prasad Bandodkar, Shelaka Gupta, Md. Imteyaz Alam and M. Ali Haider	Applied Catalysis A	66-72	560	2018	4.8	Q2
5.	Non-bonding and bonding interactions of biogenic impurities with the metal catalyst and the design of bimetallic alloys	Madhulika Gupta, Tuhin S. Khan, Shelaka Gupta, Md. Imteyaz Alam, Manish Agarwal, M. Ali Haider	Journal of Catalysis	542–556	352	2017	6.5	Q1
6.	Development of 6-amyl- α -pyrone as a potential biomass-derived platform molecule	Md. Imteyaz Alam, Shelaka Gupta, Ashish Bohre, Ejaz Ahmad, Tuhin S. Khan, Basudeb Saha and M. Ali Haider	Green Chemistry	6431	18	2016	9.2	Q1
7.	Reactivity Descriptor for Retro Diels-Alder Reaction of Partially Saturated 2-Pyrones: DFT Study on Substituents and Solvents Effect	Tuhin Khan, Shelaka Gupta, Md. Imteyaz Alam and M. Ali Haider	RSC Advances	101697-101706	6	2016	3.9	Q1
8.	On the mechanism of retro-Diels-Alder reaction of	Shelaka Gupta; Md. Imteyaz	RSC Advances	60433-60445	6	2016	3.9	Q1

	partially saturated 2-pyrones to produce bio-Renewable chemicals	Alam, Tuhin Suvra Khan; Nishant Sinha, M. Ali Haider						
9.	Mechanistic insights into thering-opening of biomass derivedlactones	Shelaka Gupta, Rishabh Arora, Nishant Sinha, Md. Imteyaz Alam and M. Ali Haider	RSC Advances	12932-12942	6	2016	3.9	Q1
10.	Bio-renewable chemicals: Feedstocks, technologies and the conflict with food production	Soubhik K.Bardhan, Shelaka Gupta, M.E.Gorman, M. Ali Haider	Renewable and Sustainable Energy Reviews	506–520	51	2015	16.3	Q1
11.	Simultaneous desulfurization and denitrogenation ofliquid fuel by nickel-modified granular activatedcarbon	Sandeep Kumar Thaligari, Shelaka Gupta, Vimal Chandra Srivastava, Basheshwer Prasad	Energy & Fuels	30	7	6161-6168	5.3	Q1

Section B: List of publications after joining IITH with no student from IITH as a co-author.

1.	Rational design of bimetallic alloys for effective hydrodechlorination of 4-chlorophenol	Chaitra S. Shenoy ^a , Shelaka Gupta*, M. Ali Haider* and Tuhin S. Khan*	New. J. Chem.	7799-7809	48	2024	2.5	Q2
2.	Oxygen Vacancy Mediated Reactivity of CaO/CuO Composite for the Synthesis of Amino-N-heterocycles	K. Rajendran, Roshni Madampadi, Sayan Shee, Rameshkumar Subramaniam, Tuhin S. Khan, Shelaka Gupta*, M. Ali Haider, Dinesh Jagadeesan*	ChemCatChem		15	2023	3.9	Q1
3.	Oxygen Vacancy-Mediated Reactivity: The Curious Case of Reduction of Nitroquinoline to Aminoquinoline by CuO	K. Rajendran, Jyoti Yadav, Tuhin S. Khan, M. Ali Haider, Shelaka Gupta*, Dinesh Jagadeesan*	J. Phys. Chem. C	8576–8584	127	2023	3.1	Q1
4.	CuO as a reactive and reusable reagent for the hydrogenation of nitroarenes	K Rajendran, N Pandurangan, CP Vinod, Tuhin S , Shelaka Gupta*, M Ali Haider, D Jagadeesan*	Applied Catalysis B: Environmental	120417 - 120427	297	2021	21.1	Q1

5.	Understanding the origin of structure sensitivity in hydrochlorination on of trichloroethylene on a palladium catalyst (Front Cover Page Illustration)	Chaitra S Shenoy, Tuhin S Khan, Kirti Verma, Mesfin Tsige, Kshitij C Jha, M Ali Haider, Shelaka Gupta	Reaction Chemistry & Engineering	2270-2279	6	2021	3.6	Q2
6.	Design of highly stable MgO promoted Cu/ZnO catalyst for clean methanol production through selective hydrogenation of CO ₂	Sachin Kumar Sharma, Tuhin Suvra Khan, Rajib Kumar Singh, Bappi Paul, Mukesh Kumar Poddar, Takehiko Sasaki, Ankur Bordoloi, Chanchal Samanta, Shelaka Gupta, Rajaram Bal	Catalysis A: General	9	623	2021		
7.	Effect of substituents and promoters on the Diels–Alder cycloaddition reaction in the biorenewable synthesis of trimellitic acid	Tuhin Suvra Khan, Shelaka Gupta, Maaz Ahmad, Md Imteyaz Alam, M Ali Haider	RSC Advances	30656-30670	10	2020	3.9	Q1

Section C: List of publications after joining IITH with students from IITH as a co-author.

S No.	Publication Title	Authors	Journal Name	Page No.	Journal	Year	I.F.	Quartile
1.	Elucidating the role of solvent in C=C bond hydrogenation of maleic acid at metal-solvent interface by AIMD and DFT simulations	Aashi D. Parekh, Pallavi Dandekar, Uzma Anjum, Shelaka Gupta*	Journal of Catalysis	116212	448	2025	6.5	Q1
2.	Facet dependence for solvent-modulated proton-coupled electron transfer in furfural acetalization on Pd nanostructures	Govind Porwal, Pallavi Dandekar, Twinkle Gorai, Tuhin Suvra Khan, M. Ali Haider*, Shelaka Gupta*, C.P. Vinod*	Chemical Engineering Journal	163159	514	2025	13.2	Q1
3.	Machine learning assisted approximation of descriptors (CO and OH) binding energy on Cu-based bimetallic alloys	Pallavi Dandekar, Aditya Singh Ambesh, Tuhin Suvra Khan and Shelaka Gupta*	Phys. Chem. Chem. Phys.	7151-7168	27	2025	3.7	Q1

4.	C=C Bond Hydrogenation vs C-O Bond Hydrogenolysis of Furfuryl Alcohol on Ru: A DFT Study	Aishwarya Kollamvalappil Narayanan, Devansh Kaushik, Shelaka Gupta*	The Journal of Physical Chemistry C	18265–18272	128	2024	3.1	Q1
5.	Property tunable two-dimensional zinc sulfoselenides as photocatalysts for enhanced hydrogen peroxide production: Experimental and DFT analyses	Akshay Tikoo, Shelaka Gupta*, Praveen Meduri*	Journal of Catalysis	115684	438	2024	6.5	Q1

7. List of patents:

Patents after joining IITH with IITH students as co-inventer

S.No.	Patent Title	Name of Applicant(s)	Patent No.	Award Date	Agency/Country	Status
1.	Conversion of biomass-derived 6-amyl- α -pyrone (6-PP) to n-decane”	Aishwarya K.N., S. Gupta, Md. I. Alam, M.A. Haider , Debaprasad Shee	566760	27/5/2025	Indian Patent Application	Awarded

Patents before joining IITH

S.No.	Patent Title	Name of Applicant(s)	Patent No.	Award Date	Agency/Country	Status
1.	A process for the production of unsaturated c-9 linear ketones	M. A. Haider, S. Gupta, E. Ahmad, Md. I. Alam	310920	05/04/2019	Indian Patent Application	Awarded
2.	Process for preparation of hydric alcohols	M. A. Haider, Md. I. Alam, S. Gupta, B. Saha	316774	25/07/2019	Indian Patent Application	Awarded
3.	Process for Producing δ -decalactone (DDL)	M. A. Haider, Md. I. Alam, S. Gupta, A. Narang, Tuhin. S. Khan, Varsha Kumari	(Application No) 201711037804	-	Indian Patent Application	Filed on 25/10/2017

8. Books/Reports/Chapters/General articles etc.

After Joining IITH

S.No.	Title	Author Name	Publisher	Year of Publication
1.	Understanding Biomass Chemistry Using Multiscale Molecular Modeling Approach	Shelaka Gupta	Springer Nature	2021
2.	Hydrodeoxygenation of Lignin-Derived Platform Chemicals on Transition Metal Catalysts	Shelaka Gupta and M. Ali Haider	Elsevier	2022

Before Joining IITH

S.No.	Title	Author Name	Publisher	Year of Publication
1.	Biological Routes for the Synthesis of Platform Chemicals from Biomass Feedstocks	Md Imteyaz Alam, Shelaka Gupta, Ejaz Ahmad, M. Ali Haider	Springer	2017
2.	Integrated Bio- and Chemocatalytic Processing for Bio-renewable Chemicals and Fuels	Md Imteyaz Alam, Shelaka Gupta, Ejaz Ahmad, M. Ali Haider	Elsevier	2015

9. Invited Talks:

1. Rational Design of Transition Metal Catalysts to Produce Long Chain Hydrocarbons from Biomass, PCCP Symposium Sphere 2025, IISC Bangalore, Oct 16-18, 2025.
2. Towards rational design of transition metal catalysts for the upgradation of biorenewables, CACEE, TIFR Mumbai, December 16-20, 2024.
3. Theory Guided Design of Catalytic Materials for the Sustainable Production of Fuels and Chemicals, Catalysis and Reaction Engineering Symposium (CARES) 2024, Indian Institute of Technology Madras, September 26-28, 2024.
4. Elucidating the facet dependence for solvent-mediated furfural acetalization reaction on Pd nanostructures, Carbon Management in Chemical Industry, Indo-German Bilateral Workshop, BASF, Mumbai, India, 27-29th Feb. 2024.
5. An Introduction to *ab-initio* DFT Simulations for Heterogeneous Systems Training on Computational Materials Engineering, Department of Metallurgical and Materials Engineering, MNIT Jaipur, 22nd to 27th January 2024
6. Proton Coupled Electron Transfer in Furfural Acetylation Reaction on Pd Nanostructures: A Combined Experimental and Theoretical Study, HPC Research Week, IIT Madras, 22nd Nov. 2023
7. Calculation of rate of chemical and biological reactions using transition state theory, TEQIP-III Short Term Course on Basic Principles of Density Functional Theory and Molecular Dynamic Simulations, MNIT Jaipur, 25th Nov. 2020
8. Design of transition metal based heterogeneous catalyst for the production of biorenewables. 13th August 2019, HPC, IIT Delhi.

10. Conferences:

1. Production and valorization of bio-based platform molecules for future bio-refinery. Catalysis for clean energy and sustainable future. 31st May-1st June 2019, IIT Delhi.
2. Rational Design of Terrace, Step and Corner Sites of the Metal or Bimetallic Catalysts for Hydrogenation and Hydrodeoxygenation of Biomass-Derived Platform Molecules. 2018 MRS Fall Meeting and Exhibit, Boston, Massachusetts, U.S.A., November 2018.
3. Catalytic conversion of biomass-derived platform molecules: mechanistic insights, fundamental challenges and opportunities for rational catalyst design, Science Day 2018, IIT Delhi, India.
4. Rational design of Metal or Bimetallic Catalysts for Hydrogenation and Hydrodeoxygenation of Biomass-derived Platform Molecules. Bioprocessing India (2018), Delhi, India.

5. DFT study on the substituent and solvent effect governing ring-opening and decarboxylation of biomass-derived cyclic esters. APCAT, Mumbai, India, January 2017.
6. Valorization of biomass-derived 6-amyl- α -pyrone into fuels and chemicals, North American Catalysis Society Meeting 2017 (NAM25), Denver, Colorado, USA, June, 2017. Ring-opening and decarboxylation of biomass derived lactones and pyrones. International Symposia on Chemical Reaction Engineering (ISCRE 24), Minneapolis Minnesota, U.S.A., June 2016.
7. Catalytic processing of 2-pyrones to produce bio-renewable chemicals. HETIS, Chandigarh, India, 2014.
8. Insights into the ring opening and decarboxylation of biomass derived cyclic esters. Faraday Discussion, London, 2016.
9. Combined bio- and chemo-catalytic conversion of biomass derived lactones into chemicals. AOC-5 2015, Royal Society of Chemistry & Green Chemistry, New Delhi, India.
10. A density functional theory study on the ring opening mechanism of biomass derived 2- pyrone molecules. Bioprocessing India 2014, Mumbai, India.

11. Teaching Summary after Joining IITH:

S. No.	Course Number	Course Title	Year-Semester	Number of Credits	Number of students	Feedback obtained
1.	CH5050	Heterogeneous Catalytic Reaction Engineering	AUG20-NOV20	3	18	4.38
2.	CH1130	Chemical Process Calculations	MAR21-JUN21	2	36	Not able to see
3.	CH5190	Molecular Modelling of Catalytic Reactions	JAN21-APR21	3	7	Not able to see
3.	CH1130	Chemical Process Calculations	MAR22-JUN22	2	42	4.47
4.	CH5050	Heterogeneous Catalytic Reaction Engineering	AUG22-NOV22	3	31	4.62
5.	CH5190	Molecular Modelling of Catalytic Reactions	JAN23-MAY23	3	13	4.87
6.	CH1130	Chemical Process Calculations	MAR23-JUN23	2	42	Not able to see
7.	CH5050	Heterogeneous Catalytic Reaction Engineering	AUG23-NOV23	3	31	4.41
8.	CH5190	Molecular Modelling of Catalytic Reactions	JAN24-MAY24	3	6	4.80
9.	CH5050	Heterogeneous Catalytic Reaction Engineering	AUG24-NOV24	3	23	4.54
10.	CH5190	Molecular Modelling of Catalytic Reactions	JAN25-MAY25	3	10	Not able to see
11.	CH3110	Mass Transfer 1	AUG25-NOV 25	3	44	Running

12. PhD guidance: Number of completed and ongoing PhD students at IITH. 5

1. Akshay Tikoo (CH18M20P000003, co-supervised by Dr. Praveen Meduri) **Viva Completed**
2. Pallavi Dandekar (CH21RESCH01005, PMRF Fellow) **OC completed**
3. Aishwarya K N (CH22RESCH11007) **Ongoing**
4. Subham Das (ID22RESCH11009) with Dr. Mayur Vaidya (ID student) **Ongoing**
5. Ramya S Lakshmi (ID24RESCH11008) with Dr. Mayur Vaidya (ID student) **Ongoing**

13. Projects (sponsored/Consultancy) in the following format handled at IITH:

Sl. No.	As PI or Co-PI	Project Title	Sponsoring Agency	Period of project	Project Cost (in Rs., in Lakhs)
1.	PI	Rational Design of Transition Metal Catalysts to Produce Value-Added Chemicals from Biomass Derived 2-pyrone Molecules. (2022-24)	SERB	2 years	Rs. 2891250
2.	Co-PI	A combined experimental and theoretical approach towards rational design of supported metal catalyst for the reductive depolymerization of corncob lignin to produce bulk aromatic chemicals (2021-2025)	SERB	3 years	Rs. 40,18,660
3.	PI	Rational Design of Catalyst for the Catalytic Transfer Hydrogenation of Biomass-Derived (2020-2022)	IITH	2 years	Rs. 25,00,000

14. Administrative Activities at IITH.

- Warden of Aryabhata and Bhaskara Hostel (Since March 2023)
- DPGC Member CHE Department (Till Nov. 2023)
- DUGC Member CHE Department (From Nov. 2023 to September 2025)
- Faculty Search Committee Member (Since Dec. 2024)
- Convenor of Outreach Cell CHE Department (Since Nov. 2023)
- Member of Student Grievance Cell (Since April 2024)